

Basics

Logic: Logic is the discipline that considers the methods of reasoning. It provides the rules and techniques for determining whether an argument is valid or not.

In everyday life, we use reasoning to prove different points. For example: to prove our parents that we passed an exam, we might show the test and the score.

Similarly, in mathematics and computer science, mathematical logic or logic is used to prove results. To be specific, in mathematics we use logic and logical reasoning to prove the correctness of programs and also to prove the theorems.

Statement or proposition:

A statement, or a proposition is the declarative sentence that is either true or false but not both and one of the values 'truth' or 'falsity' that is assigned to a statement is called its truth values.

We abbreviate 'truth' to **T** or **1** and 'false' to **F** or **0**.

Examples:

1. Canada is a country.
2. 5 is less than 3.
3. This statement is false.
4. $1+101=110$
5. Close the door.

Connectives:

In our everyday language, we use connectives such as ‘and’, ‘but’, ‘or’ etc. to combine two or more statements to form other statements. We will discuss some of the connectives:

1. Negation (Not or $\sim$):

The negation of a statement ‘**P**’ is written as $\sim\mathbf{P}$ and read as ‘**not P**’.

The truth table for the negation is

P	$\sim\mathbf{P}$
T	F
F	T

Remark: Also we can read as ‘**it is not the case that P**’. And it is a unary operation.

Examples:

1. **P**: London is a city.

~P: London is not a city.

Or

~P: It is not the case that London is a city.

2. **P**: John is a rich man

~P: John is not a rich man

Or

~P: It is not the case that John is a rich man

2. Conjunction (And, $\wedge$):

The conjunction of two statements ' P ' and ' Q ' is written as $P \wedge Q$ and is read as ' P and Q '.

It has the truth value '**true**' only when both P and Q are true otherwise '**false**'.

Truth table:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Remark: Conjunction is a binary operation and it is symmetric.

Examples:

1. **P**: It is cold.

Q: It is rainy.

So

P $\wedge$ **Q** : It is cold and it is rainy.

2. **P**: John is a rich man

Q: John is intelligent

So

P $\wedge$ **Q** : John is a rich man and John is intelligent.

3. Disjunction (Or, 'v'):

The disjunction of two statements '**P**' and '**Q**' is written as **P v Q** and is read as '**P or Q**'.

It has the truth value '**false F**' only when both **P** and **Q** have the truth value false otherwise True.

Truth table:

P	Q	P v Q
T	T	T
T	F	T
F	T	T
F	F	F

Remark: Disjunction is also a binary operation and it is symmetric.

Examples:

1. **P**: It is cold.

Q: It is rainy.

So

P $\vee$ **Q** : It is cold or it is rainy.

2. **P**: John is a rich man

Q: John is intelligent

So

P $\vee$ **Q** : John is a rich man or John is intelligent.

4. Conditional statements ($\rightarrow$):

Let ' P ' and ' Q ' be any two statements. Then the statement ' $P \rightarrow Q$ ' is read as 'If P then Q ' and has the truth value false only when P is true and Q is false.

Truth table:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Example:

P : There is a flood.

Q : The crop will be destroyed.

So

$P \rightarrow Q$: If there is a flood then crop will be destroyed.

Remark: $P \rightarrow Q$ can be represented by anyone of the following expression:

- a) Q is necessary for P.
- b) P is sufficient for Q
- c) P only if Q.

The statement P is called as the antecedent and Q the consequent in $P \rightarrow Q$.

Again, according to the definition, it is not necessary that there be any kind of relation between P and Q in order to form $P \rightarrow Q$.

Example:

P: The Sun is shining today.

Q: $2+7 > 4$.

So

$P \rightarrow Q$: If the Sun is shining today then $2+7 > 4$.

5. Bi-conditional statements ($\leftrightarrow$):

Let 'P' and 'Q' be any two statements. Then the statement ' $P \leftrightarrow Q$ ' is read as '**P if and only if Q**' or '**P iff Q**' and has the truth value true if both P and Q are identical.

Truth table:

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Example:

P: x is an even number.

Q: x is divisible by 2.

So

$P \leftrightarrow Q$: x is an even number if and only if x is divisible by 2.

Atomic Statements:

Declarative sentences which cannot be further spilt into simple sentences are called as atomic statements (or primary statements or primitive statements).

Example: p is a prime number.

Compound statements:

New statements can be formed from atomic statements by using connectives 'and' , 'but' , 'or' etc., the resulting statements are called as molecular or compound statements.

Statement formula:

If we represent the definite statements by means of the statement variables like P , Q , R ,..., then the compound statement is called statement formula.

Example:

If P: Martin is poor

Q: Martin is happy

Write the following statements in symbolic form

- (a) Martin is rich and happy
- (b) Martin is neither poor nor happy
- (c) Martin is rich or he is both poor and unhappy.

Solution:

(a) $\sim P \wedge Q$

(b) $\sim P \wedge \sim Q$

(c) $\sim P \vee (P \wedge \sim Q)$

Converse:

The statement $Q \rightarrow P$ is called the converse of the implication $P \rightarrow Q$.

Example:

P: Today is Sunday.

Q: I will go for a walk.

So,

$Q \rightarrow P$: If I will go for a walk then today is Sunday.

Inverse:

The statement $\sim P \rightarrow \sim Q$ is called the inverse of the implication $P \rightarrow Q$.

$\sim P \rightarrow \sim Q$: If today is not Sunday then I will not go for a walk.

Contrapositive:

The statement $\sim Q \rightarrow \sim P$ is called the contrapositive of the implication $P \rightarrow Q$.

Example:

P: Today is Sunday.

Q: I will go for a walk.

So,

$\sim Q \rightarrow \sim P$: If I will not go for a walk then today is not Sunday.

Truth table for statement formula

$$S = (P \rightarrow Q) \vee (\sim P \wedge Q)$$

P	Q	$\sim P$	$P \rightarrow Q$	$\sim P \wedge Q$	S
T	T	F	T	F	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	F	T

Construct the truth table for the following statement formulae

(i) $(p \rightarrow q) \longleftrightarrow (\neg p \vee q)$

(ii) $p \wedge (p \vee q)$

(iii) $(p \rightarrow q) \rightarrow p$

(iv) $\neg(p \wedge q) \longleftrightarrow (\neg p \vee \neg q)$

(v) $(p \vee \neg q) \rightarrow q$

$$1. S = (Q \wedge (P \rightarrow Q)) \rightarrow P$$

$$2. S = (P \vee Q) \wedge (\sim R \leftrightarrow Q)$$

Tautology: A statement formula which is true regardless of the truth values of the statements which replace the variables in it is called a universally valid formula or a Tautology. It is denoted by T.

Example: $P \vee \sim P$

P	$\sim P$	$P \vee \sim P$
T	F	T
F	T	T

Contradiction: A statement formula which is false regardless of the truth values of the statements which replace the variables in it is called a contradiction. It is denoted by F.

Example: $P \wedge \sim P$

P	$\sim P$	$P \wedge \sim P$
T	F	F
F	T	F

Contingency: A statement formula which is neither a tautology nor a contradiction is called a contingency.

Example: $P \vee Q$

Examples: Using truth table prove that the following statement formula is a Tautology

$$(P \wedge Q) \vee (P \rightarrow R) \vee (\sim Q \rightarrow \sim R)$$

P	Q	R	$P \wedge Q$ (1)	$P \rightarrow R$ (2)	$(1) \vee (2)$ (3)	$\sim Q$	$\sim R$	$\sim Q \rightarrow \sim R$ (4)	$(3) \vee (4)$
T	T	T	T	T	T	F	F	T	T
T	T	F	T	F	T	F	T	T	T
T	F	T	F	T	T	T	F	F	T
T	F	F	F	F	F	T	T	T	T
F	T	T	F	T	T	F	F	T	T
F	T	F	F	T	T	F	T	T	T
F	F	T	F	T	T	T	F	F	T
F	F	F	F	T	T	T	T	T	T

Equivalence of formulas: Let A and B be two statement formulas and let $P_1, P_2, P_3, \dots, P_n$ denote all the variables occurring in both A and B . If the truth value of A is equal to the truth value of B for every 2^n possible combination of truth values, then A and B are said to be equivalent and it is denoted by $A \Leftrightarrow B$

Example: $\sim\sim P \Leftrightarrow P$, $P \vee P$, etc.

Remarks:

1. $\Leftrightarrow$ is not a connective.
2. Two statement formulas A and B are equivalent iff $A \leftrightarrow B$ is a tautology.

Example : Prove that

(a) $P \rightarrow Q \Leftrightarrow \sim P \vee Q,$

(b) $P \leftrightarrow Q \Leftrightarrow (P \rightarrow Q) \wedge (Q \rightarrow P)$

(c) $\sim (P \wedge Q) \Leftrightarrow \sim P \vee \sim Q$

(d) $\sim (P \vee Q) \Leftrightarrow \sim P \wedge \sim Q$

v

Equivalent formulae:

1. $P \vee P \Leftrightarrow P$ and $P \wedge P \Leftrightarrow P$ (Idempotent law)
2. $(P \vee Q) \vee R \Leftrightarrow P \vee (Q \vee R)$ and $(P \wedge Q) \wedge R \Leftrightarrow P \wedge (Q \wedge R)$ (Associative law)
3. $P \vee Q \Leftrightarrow Q \vee P$ and $P \wedge Q \Leftrightarrow Q \wedge P$ (Commutative law)
4. $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$ and $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$ (Distributive law)